NUMBER SYSTEM MADE EASY | VIDEO 5

Binary ↔ Hexadecimal

Grouping into Four Bits



Quick Review

Binary

Base 2

Hexadecimal

Base 16

Hex digits: 0–9 and A–F

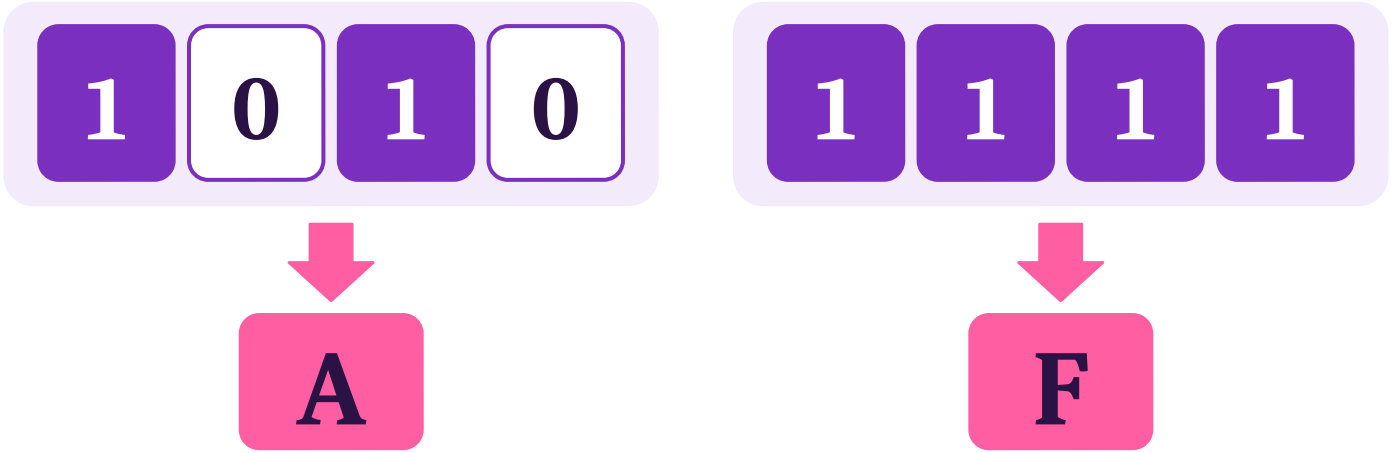
A	B	C	D	E	F
= 10	= 11	= 12	= 13	= 14	= 15

Binary → Hexadecimal Table

0000 → 0	0001 → 1	0010 → 2	0011 → 3
0100 → 4	0101 → 5	0110 → 6	0111 → 7
1000 → 8	1001 → 9	1010 → A	1011 → B
1100 → C	1101 → D	1110 → E	1111 → F

Binary → Hex: 10101111_2

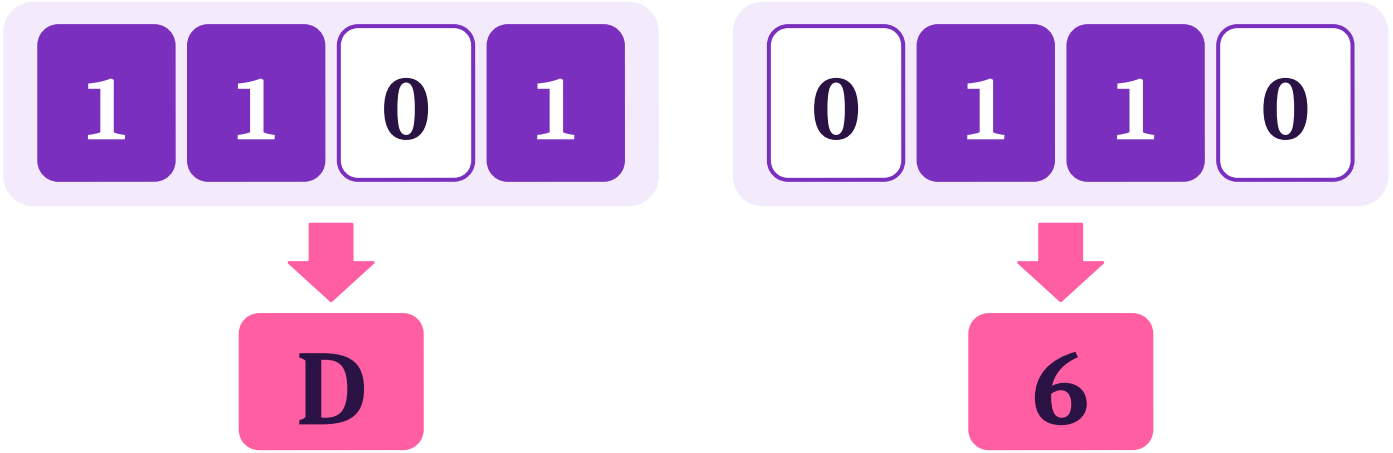
1 Group into fours | 2 Group → hex digit | 3 Join the digits



Answer: $10101111_2 = AF_{16}$

Binary → Hex: 11010110_2

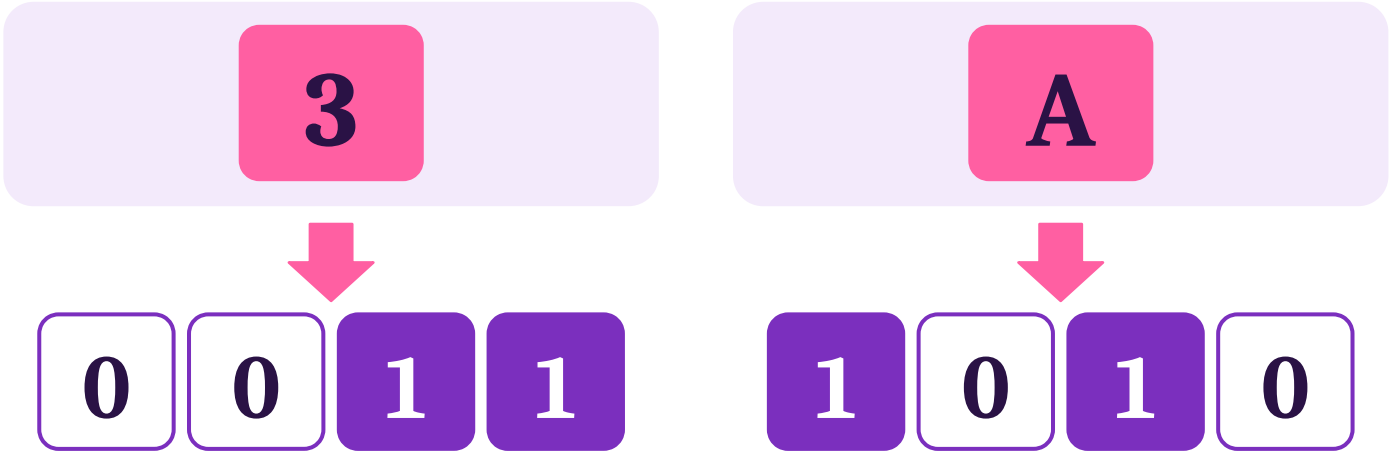
1 Group into fours | 2 Group → hex digit | 3 Join the digits



Answer: $11010110_2 = D6_{16}$

Hex → Binary: $3A_{16}$

1 Split into digits | 2 Digit → 4 bits | 3 Join the bits



Answer: $3A_{16} = 00111010_2$

Your Turn!

Convert:

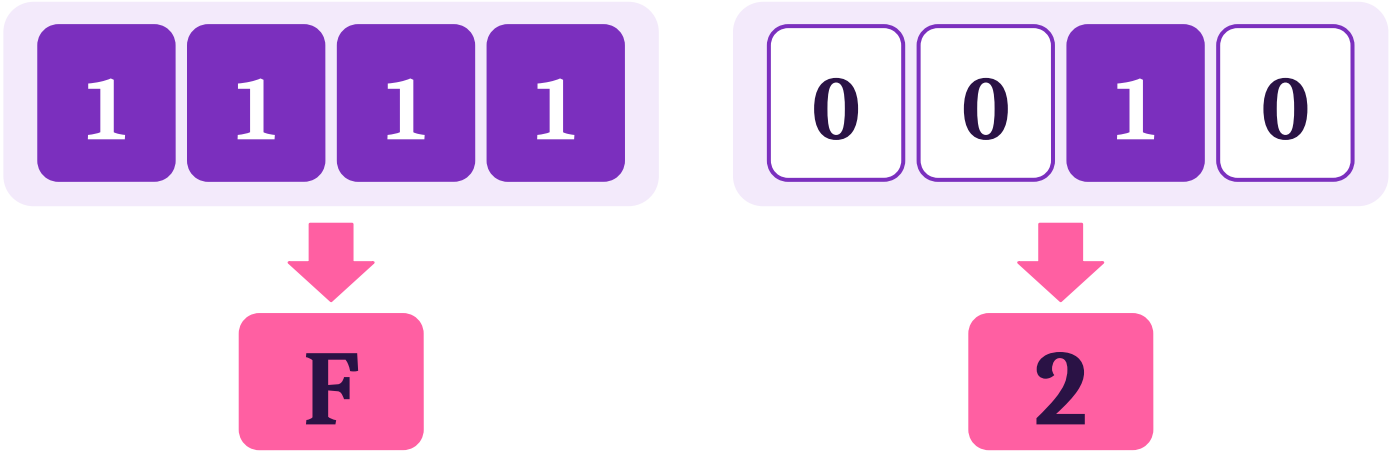
$$11110010_2 \rightarrow ?_{16}$$

You have 20 seconds!



Answer: 11110010_2 in Hex

1 Group into fours | 2 Group → hex digit | 3 Join the digits



Answer: $11110010_2 = F2_{16}$

Recap

Binary → Hexadecimal

Group 4

Hexadecimal → Binary

4 bits per digit

Next video: Decimal → Octal